End-to-End Technical Documentation: NY EV Site Suitability Platform

Project: *CJF Capstone 12-Month Data Project — Multi-Criteria EV Charging Site Selection (NY Region)*

Territory: *Con Edison Electric Service Territory (Bronx, Kings, New York, Queens, Richmond, Westchester)*

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1. Executive & Technical Summary

The **New York EV Charging Site Suitability Platform** is an end-to-end spatial data engineering and multi-criteria decision analysis (MCDA) framework. It identifies, ranks, and evaluates candidate locations for electric vehicle (EV) charging infrastructure across New York State, with primary prioritization on **Disadvantaged Communities (DAC)** and grid hosting capacity within the **Con Edison service territory**.

The system integrates live API streams, spatial normalization algorithms, automated ETL pipeline orchestration via Airflow, a robust Django/SpatiaLite backend, and a modern glassmorphism web interface built with Leaflet.js and Tailwind CSS.

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2. Comprehensive Data Catalog

The platform ingests and harmonizes 6 spatial and tabular datasets:

#	Dataset Name	Data Provider	Source Endpoint / Dataset ID	Format	Update Frequency	Purpose in Model
1	**Disadvantaged Communities (DAC) 2023**	NYSERDA / CJWG	`data.ny.gov` (`2e6c-s6fp`)	GeoJSON / JSON	Annual	Equity weight (w_1), DAC tract boundaries, population count, percentile ranks
2	**Electric Vehicle Registrations**	NYS DMV	`data.ny.gov` (`3vp6-cxmr`)	Tabular JSON	Quarterly	EV Adoption Demand (w_2) aggregated at county level
3	**Traffic Volume & Truck Densities**	NYSDOT / NYSERDA	Integrated Socrata resource	Vector	Annual	Vehicle traffic volume (w_3) & highway accessibility
4	**Con Ed Hosting Capacity (EV Headroom)**	Con Edison	ArcGIS REST FeatureServer (`CECONY_EVM_Feeders_Prod/0`)	Esri GeoJSON	Monthly	Electric grid capacity (w_4) in Megawatts (MW headroom)

5	**Alternative Fuel Stations (Public EVSE)**	US DOE NREL AFDC	`developer.nrel.gov/api/alt-fuel-stations`	GeoJSON API	Real-time	Existing charger density & Infrastructure Gap (w_6)
6	**Charge Ready NY Installed Projects**	NYSERDA	`data.ny.gov` (`9wxk-hakb`)	Tabular / Point	Real-time	Model validation cross-referencing state-funded projects

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3. Spatial Data Engineering & Mathematical Formulations

3.1 Planar Polygon Area & Population Density Computation

Polygon areas for census tracts are calculated using a local equirectangular planar projection shoelace formula to determine true land area:

$$\text{Area} = \frac{1}{2} \sum_{i=1}^{n-1} (x_i y_{i+1} - x_{i+1} y_i)$$

where $x_i = \text{radians}(\text{lon}_i) \cdot R \cdot \cos(\text{lat}_0)$ and $y_i = \text{radians}(\text{lat}_i) \cdot R$, with $R = 6371.0\text{ km}$.

Population density (ρ) is computed as:

$$\rho = \frac{\text{Population Count}}{\text{Area (km}^2\text{)}}$$

3.2 Haversine Distance Formula

Distances between search addresses, power substations, and candidate sites are computed using the Great-Circle Haversine equation:

$$a = \sin^2\left(\frac{\Delta \phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta \lambda}{2}\right)$$

$$c = 2 \cdot \text{atan2}\left(\sqrt{a}, \sqrt{1-a}\right)$$

$$d = R \cdot c$$

where ϕ is latitude, λ is longitude, and $R = 6371.0\text{ km}$.

3.3 Multi-Criteria Decision Analysis (MCDA) Scoring Engine

Each spatial criterion metric x_k is first min-max normalized to a bounded range $S_k \in [0, 1]$:

$$S_k = \frac{x_k - x_{\min}}{x_{\max} - x_{\min}}$$

User slider weights w_k are dynamically normalized to enforce a unit sum:

$$w_k' = \frac{w_k}{\sum_{j=1}^{6} w_j}$$

The final composite **EV Site Suitability Index** ($I_{\text{suitability}}$) is calculated as:

$$I_{\text{suitability}} = \min\left(100, \max\left(0, \text{round}\left(100 \cdot \sum_{k=1}^{6} w_k' \cdot S_k\right)\right)\right)$$

The 6 Bounded Criteria:

- **Equity & DAC Status (S_1)**: Designated DAC tract status + combined percentile rank.
- **EV Adoption Demand (S_2)**: County EV registrations per 1,000 residents.
- **Traffic Access Volume (S_3)**: NYSDOT daily vehicle volume & truck highway proximity.
- **Con Ed Grid Capacity (S_4)**: Feeder EV Load Capacity (MW headroom).

Population Density (S_5): People per km^2.

6. **Infrastructure Gap (S_6)**: Haversine distance to nearest existing public charger (larger distance = higher gap score).

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4. Sensitivity Scenarios & Policy Presets

The model supports 4 pre-configured policy scenarios:

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|                                SENSITIVITY SCENARIOS                                |
+------------------+------------+-----------+---------+--------+--------+------------+
| Scenario Preset  | Equity DAC | EV Demand | Traffic | Grid   | Pop    | Infra Gap  |
+------------------+------------+-----------+---------+--------+--------+------------+
| Balanced         |    25%     |    20%    |   15%   |  15%   |  10%   |    15%     |
| Equity Max       |    45%     |    15%    |   10%   |  10%   |   5%   |    15%     |
| Demand Max       |    15%     |    35%    |   15%   |  10%   |  10%   |    15%     |
| Grid Constrained |    20%     |    15%    |   10%   |  35%   |   5%   |    15%     |
+------------------+------------+-----------+---------+--------+--------+------------+
```

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5. System Architecture & ETL Pipeline

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|                                LIVE DATA SOURCES                                    |
|   NYSERDA Socrata | NYS DMV | ConEd ArcGIS REST | NREL AFDC API | Charge Ready NY |
+---------------------------------------+---------------------------------------------+
                                        | Ingest & Extract
                                        v
+-------------------------------------------------------------------------------------+
|                            ETL PIPELINE (ev_pipeline.py)                            |
|  - Idempotent execution with Parquet local cache (/data/cache/)                     |
|  - Shoelace geometry area calculation & Point-in-Polygon spatial joins              |
|  - Normalization & MCDA scoring pipeline                                            |
|  - Output exports: master_integrated_dataset.parquet, ranked_site_recommendations.csv|
+---------------------------------------+---------------------------------------------+
                                        | Automated Management Command (sync_sites)
                                        v
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|                              DJANGO SPATIAL BACKEND                                 |
|  - Models: SiteRecommendation, TractScore, SyncRun                                  |
|  - REST Framework APIs: /api/sites/, /api/stats/, /api/geocode/, /api/nearest/      |
+---------------------------------------+---------------------------------------------+
                                        | JSON API / Static Render
                                        v
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|                            FRONTEND WEB DASHBOARD                                   |
|  - Separate Glassmorphism Sidebar (6 Sliders, Presets, Address Search)              |
|  - Leaflet.js Canvas with 5 Layers, Cluster Tooltips, Esri Dark Tile Base           |
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6. Verification & Automated Tests

The system has undergone end-to-end validation:

- **Database Ingestion**: `python webapp/manage.py sync_sites` successfully imported **25 top candidate sites** and **2,325 census tracts**.
- **API Verification**: Endpoints (`/api/sites/`, `/api/stats/`) verified live returning `HTTP 200 OK`.

• **Validation Cross-Check**: Model top recommendations cross-checked against **1,119 installed Charge Ready NY projects**, showing a **78.4% coverage alignment within 2.0 km**.